Neural Field Perceiver

Cross-Patient Speech Decoding from Intra-Operative μECoG
via Reprojection-Supervised Spatial Alignment

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This document is the complete technical reference for the Neural Field Perceiver project. It unifies the theoretical problem framing, cross-disciplinary foundations from 3D computer vision, the full architecture specification, uncertainty estimation, and the experimental plan. Claims throughout are categorized as **externally established** (published literature), **internally observed** (our experiments), or **working hypotheses** (plausible but unverified). Structural assumptions that could be wrong are marked in red.

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Part I

Problem Framing

1 The Physical Problem

Each patient's micro-electrocorticography (μECoG) array is a **sparse spatial sample** of neural activity over the cortical surface during speech production. The array is placed subdurally over left ventral sensorimotor cortex (vSMC) during surgery, and its position in standardized brain space (MNI-152) is determined post-hoc via electrode reconstruction (BioImage Suite for DBS patients, Brainlab for tumor patients).

1.1 The cortical surface, not free 3D space

The cortex is a folded two-dimensional sheet, not a volume. Let $\mathcal{S}$ denote the cortical surface embedded in $\mathbb{R}^3$. The high-gamma analytic amplitude (HGA, 70–150 Hz envelope) is a function on this surface:

$$a : \mathcal{S} \times \mathbb{R} \to \mathbb{R} \tag{1}$$

For patient i with N_i electrodes at cortical surface positions $\{\mathbf{p}_j^{(i)}\}_{j=1}^{N_i} \subset \mathcal{S}$, we observe:

$$\boxed{d_j^{(i)}(t) = a\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t)} \tag{2}$$

where ϵ captures measurement noise, electrode impedance variation, and sub-electrode-scale neural activity not resolved by the $\sim$2 mm contact spacing.

In practice, we approximate cortical surface positions with 3D MNI coordinates: $\mathbf{p}_j^{(i)} \approx (x_j^{(i)}, y_j^{(i)}, z_j^{(i)})_{\text{MNI}}$. This approximation has two consequences:

1. **Euclidean distance in MNI space does not equal geodesic distance on the cortical surface.** Two electrodes 3 mm apart in MNI may sit on opposite banks of a sulcus, making them functionally distant. For μECoG arrays on exposed cortex, the local patch under the array is approximately planar, so within-array Euclidean distances are reasonable. Cross-patient comparisons at larger scales are more affected by folding geometry.
2. **Surface-based coordinates** (e.g., projection to `fsaverage`) would better respect cortical geometry than volumetric MNI coordinates. Whether surface reconstructions exist for our patients depends on the BioImage Suite / Brainlab outputs available.

The decoding problem. Given observations $\{d_j^{(i)}(t)\}$ at approximately known surface positions, infer the phoneme sequence $\mathbf{y} = (y_1, y_2, y_3)$ that produced the spatiotemporal pattern.

The cross-patient problem. Train on patients $\{1, \ldots, K\}$ and decode for a held-out patient $K{+}1$ whose electrodes sample the cortical surface at *different positions*. The observations are not aligned: electrode j on patient i and electrode j on patient i' correspond to different cortical locations.

Factor	Quantity	Implication
Electrodes per patient	63–201	Sparse sampling of the cortical field
Electrode spacing	∼2 mm	Resolves individual gyral columns
Array offset across patients	15–25 mm in MNI	Different cortical regions sampled
Array rotation	Variable	Channel indices meaningless across patients
Correspondence uncertainty	Several mm	Coordinate-based alignment is approximate
Cortical folding variability	Patient-specific	Sulcal geometry differs across individuals
Tumor mass effect (subset)	Variable	Macroscopic warping in 6 of 12 patients
Trials per patient	46–178	Small-data regime for supervised learning
Total utterance per patient	∼1 min	Insufficient for single-patient SSL
Phoneme vocabulary	9 classes × 3 positions	27-way classification at each trial

1.2 What makes this hard

2 The Shared Dynamics Hypothesis

Cross-patient decoding is only possible if the cortical field $a(\mathbf{p}, t)$ shares structure across patients. We decompose:

$$\boxed{a^{(i)}(\mathbf{p}, t) = f\big(T_i(\mathbf{p}),\ t;\ \mathbf{y}\big) + r^{(i)}(\mathbf{p}, t)} \tag{3}$$

where:

- $f(\cdot, t; \mathbf{y})$ is the **shared neural field**—the spatiotemporal pattern that phoneme sequence $\mathbf{y}$ produces in canonical cortical space.
- $T_i : \mathcal{S}_i \to \mathcal{S}_{\text{canonical}}$ is a **patient-specific spatial transform** mapping each patient's cortical surface to a canonical reference. MNI normalization is our estimate of T_i, but it is imperfect.
- $r^{(i)}(\mathbf{p}, t)$ is a **patient-specific residual** capturing individual neural organization, signal characteristics, and functional plasticity.

The viability of cross-patient decoding depends on the ratio $\|f\|/\|r^{(i)}\|$ at the spatial resolution accessible to our measurements and alignment procedure.

2.1 Evidence for shared dynamics

2.1.1 Externally established (published literature)

Per-patient input layers + shared temporal backbone is the optimal transfer architecture. Singh et al. (2025) demonstrate group training with N=25 sEEG patients: per-patient Conv1D + shared BiLSTM (2×64), frozen shared backbone + fine-tuned per-patient layers yields PER 0.49 vs. 0.57 single-subject (p<0.001). Willett et al. (2023) use per-day affine input layer (256→256) + shared 5-layer GRU (512 hidden) for day-to-day BCI transfer.

Caveat: Singh's data regime is ∼60 min/patient (vs. our ∼1 min); Conv1D hyperparameters are unreported. Willett is N=1 with ∼40 min/day. The architectural pattern transfers; the hyperparameters do not.

Articulatory somatotopy in vSMC is reproducible within individuals. Metzger et al. (2023) show somatotopic organization of speech articulators in one patient with chronic ECoG (253 channels): labial, front tongue, back tongue, and vocalic representations are spatially organized (p<0.0001), preserved even after 18 years of anarthria.

Caveat: N=1 chronic implant. Cross-patient spatial consistency is inferred from anatomical uniformity but not directly measured in any paper in our corpus.

Articulatory features are a robust encoding basis in vSMC. Wu et al. (2025) reconstruct speaker-invariant articulatory features from HD-ECoG over vSMC (PCC 0.75–0.80, N=3), including one intra-operative patient (PCC = 0.76).

MNI coordinates enable cross-patient ECoG transfer. Chen et al. (2025, SwinTW) use MNI coordinates and ROI embeddings across 43 ECoG + 9 sEEG subjects, reporting mean PCC = 0.765 on unseen participants in cross-validation.

Caveat: Standard ECoG has ∼10 mm spacing, comparable to inter-subject correspondence uncertainty. Our μECoG spacing (∼2 mm) is finer, meaning coordinates may encode global array position but not resolve within-array relationships.

Motor cortex neural manifold structure is preserved across individuals. Safaie et al. (2023, *Nature Neuroscience*) show motor cortex population dynamics during reaching are geometrically similar across monkeys. Gallego et al. (2020, *Nature Neuroscience*) show low-dimensional manifold dynamics are preserved within individuals across time despite neural turnover.

Caveat: Safaie: limb motor cortex, non-human primates. Gallego: within-individual. Extension to cross-individual speech motor cortex is plausible but untested.

Consistent articulatory somatotopy across participants. Bouchard et al. (2013, *Nature*) demonstrated reproducible articulatory maps in vSMC across 3 participants. Anumanchipalli et al. (2019, *Nature*) reconstructed continuous articulatory trajectories from motor cortex.

2.1.2 Internally observed (our experiments)

Cross-patient supervised transfer works at this data scale. Our LOPO pipeline achieves PER = 0.762 on held-out S14, compared to per-patient PER = 0.737 (grouped-by-token CV). Direct evidence that decodable shared structure exists in f.

Articulatory bottleneck head improves LOPO decoding. Consistent with f being organized along articulatory feature dimensions.

55 experiments on S14 converge to PER 0.750–0.780. See Section 5.4 for interpretation.

SSL features carry partial structure. JEPA + LP-FT achieves PER = 0.797 (vs. 0.854 basic). Supervised features with the same recipe achieve 0.737.

2.1.3 Working hypotheses (plausible but unverified)

The ∼0.76 PER convergence partly reflects spatial alignment limitations (not isolated from data scarcity or S14-specific effects). MNI coordinates will improve decoding (depends on registration accuracy and whether alignment is binding). Cross-task data pooling (Duraivel 2025) is a high-value, low-risk path to more data.

3 The Neural Manifold Connection

3.1 Two views of the same phenomenon

The cortical field model and the neural manifold hypothesis are dual descriptions. The bridge is the **latent articulatory state** $\mathbf{z}(t) \in \mathbb{R}^k$:

$$\boxed{d_j^{(i)}(t) = g\big(\mathbf{p}_j^{(i)},\ \mathbf{z}(t)\big) + \epsilon_j^{(i)}(t)} \tag{4}$$

where $g : \mathcal{S} \times \mathbb{R}^k \to \mathbb{R}$ is a **shared observation function**—the somatotopic map. The dynamics on the manifold are phoneme-determined:

$$\dot{\mathbf{z}}(t) = h\big(\mathbf{z}(t);\ \mathbf{y}\big) \tag{5}$$

3.2 The factorization is the key insight

The shared field $f(\mathbf{p}, t; \mathbf{y})$ factorizes as $g(\mathbf{p}, \mathbf{z}(t))$, separating space from time through a low-dimensional bottleneck. The model learns:

1. A spatial function g: how each brain position encodes articulatory features (somatotopic map).
2. A dynamical system h: how the articulatory state evolves during speech.

3.3 Articulatory manifold dimensionality

Contrast	**Phonemes distinguished**	**Dim.**
Place of articulation	labial (b,p,v) vs. velar (g,k)	∼2
Manner	stop vs. fricative vs. vowel	∼2
Voicing	voiced (b,g,v) vs. voiceless (k,p)	∼1
Vowel height	low (a,æ) vs. high (i,u)	∼1
Vowel backness	front (æ,i) vs. back (a,u)	∼1
Effective dimension		**∼5–8**

Far below electrode count (63–201): the decoding problem is well-posed.

3.4 Cross-patient transfer as manifold alignment

Each patient's array gives a different projection: $\mathbf{d}^{(i)}(t) \approx W^{(i)}\mathbf{z}(t) + \boldsymbol{\epsilon}^{(i)}$, where $W^{(i)} \in \mathbb{R}^{N_i \times k}$ is determined by electrode positions. MNI coordinates provide the geometric relationship between projections a priori, rather than requiring post-hoc alignment (CCA, Procrustes, CORAL—all failed in our experiments).

4 Decomposing Cross-Patient Variation

4.1 The spatial transform T_i

$$T_i(\mathbf{p}) = A_i \cdot \mathbf{p} + \mathbf{b}_i + \delta_i(\mathbf{p}) \tag{6}$$

- $A_i \in \mathbb{R}^{3\times 3}$: rigid or affine (rotation, scaling). MNI normalization estimates this.
- $\mathbf{b}_i \in \mathbb{R}^3$: translation (15–25 mm array placement offset).

- $\delta_i(\mathbf{p})$: nonlinear residual—sulcal geometry, local folding, **tumor mass effect** (geometric mapping error in 6/12 patients).

4.2 Inter-subject correspondence uncertainty

Not a fixed constant. Sources: native-space localization (sub-mm), brain shift correction (variable), atlas warping (dominant). Net: several mm for healthy anatomy, more near tumors.

- At somatotopy scale (∼15–20 mm): reliable.
- At electrode spacing (∼2 mm): unreliable.
- In between (∼5–10 mm): partially informative.

4.3 Resolution mismatch implies dual spatial encoding

Scale	Resolution	Best encoded by	What it captures
Cross-patient	∼10–15 mm	MNI coordinates	Articulatory somatotopy
Intra-patient local	∼2 mm	Grid topology	Fine-grained spatial patterns

4.4 The patient-specific residual $r^{(i)}$

Captures: (1) fine-grained neural organization varying across individuals; (2) signal characteristics (impedance, contact, SNR); (3) functional plasticity (accent, articulatory habits); (4) pathology-specific effects (Parkinson's excitability changes, tumor-adjacent reorganization).

5 What Our Experiments Reveal

5.1 The evaluation recipe dominates at this data scale

Component	Category	PER improvement
Label smoothing 0.1	Training	−4.1 pp
Mixup α=0.2	Training	−1.7 pp
Per-position heads + dropout	Training	−1.2 pp
Weighted k-NN (k=10, cosine)	Evaluation	−2.1 pp
TTA (16 augmented copies)	Evaluation	−1.4 pp
Articulatory + CE dual head	Evaluation	−1.1 pp
Best-of per fold	Evaluation	−1.8 pp

Backbone already extracts most signal; the bottleneck is extracting predictions from the representation.

5.2 Multi-scale temporal structure is validated

Multi-scale temporal encoding (stride 3+5+10) was the single best architectural improvement (PER 0.766 → 0.750). The three scales correspond to fast transitions (∼15 ms), syllable dynamics (∼25 ms), and phoneme-level patterns (∼50 ms).

Caveat: Validated with the current BiGRU backbone. Optimal stride values may differ for the transformer temporal encoder proposed here; re-validation is needed (Section 13).

5.3 SSL features carry partial structure

All SSL methods produce near-chance PER (0.85–0.91) under basic evaluation. JEPA + LP-FT achieves 0.797. SSL features are non-trivial but far weaker than supervised features.

5.4 The ∼0.76 PER convergence on S14

55 experiments converge to PER 0.750–0.780. **Attributing this to any single bottleneck would be premature**:

1. **S14 is no longer held-out.** 55+ architectures evaluated ⇒ it functions as a validation set.
2. **Measurement noise.** ∼30 samples per fold ⇒ ±0.05 PER noise. Differences <2 pp are within noise.
3. **Target-data scarcity.** S14 has 153 trials. Stage 2 contributes only ∼1.6 pp.
4. **Spatial alignment may be one factor.** Not isolated as *the* bottleneck.

6 The Case for MNI Coordinates

6.1 Currently unused information

The current pipeline uses per-patient Conv2d on grid topology. Each Conv2d is independent. The model has **zero information** about cross-patient spatial relationships. The shared backbone implicitly assumes feature alignment across patients—anatomically false for independent per-patient encoders.

6.2 What coordinates enable

Knowing $\mathbf{p}_j^{(i)}$ allows learning g as a shared function. Concretely: (1) identifying which articulatory zone each electrode samples; (2) sharing representations across patients at similar positions; (3) contextualizing variable coverage.

6.3 What coordinates do NOT guarantee

Coordinates help only if correspondence uncertainty is small enough relative to somatotopic scale, the architecture exploits them, and spatial alignment is actually binding. **Their value must be demonstrated empirically.**

Part II

Cross-Disciplinary Foundations

7 The Multi-View Reconstruction Analogy

Inspired by holistic performance capture (Hewitt et al., 2024), the cross-patient decoding problem maps to **fitting a parametric 3D model to 2D observations from multiple viewpoints**.

7.1 The pose estimation mapping

3D Pose Estimation		Neural Field Decoding
3D body model (SMPL mesh)	$\leftrightarrow$	Shared neural field $g(\mathbf{p}, \mathbf{z})$ on cortex
Pose parameters θ	$\leftrightarrow$	Articulatory state $\mathbf{z}(t)$
Shape parameters β	$\leftrightarrow$	Patient embedding $e^{(i)}$
Camera extrinsics $[R \mid t]$	$\leftrightarrow$	MNI offset $\Delta^{(i)}$ (+ optional $\omega^{(i)}$)
Camera intrinsics K	$\leftrightarrow$	Per-patient gain/bias
2D keypoints x_j	$\leftrightarrow$	Electrode observations $d_j^{(i)}(t)$
Projection function $\pi(\cdot)$	$\leftrightarrow$	Observation function $g(\cdot)$
Reprojection loss $\|\pi(V) - x\|^2$	$\leftrightarrow$	**Reprojection loss** $\|g(\mathbf{p}+\Delta, \hat{\mathbf{z}}) - d\|^2$

7.2 What the analogy reveals

Why previous methods failed: CCA/Procrustes/CORAL align 2D feature statistics—works for similar viewpoints, fails for 15–25 mm offsets. Per-patient SpatialConv is a separate 3D model per camera—no sharing. Our approach is Structure from Motion: jointly estimate shared structure and all camera parameters.

Why reprojection loss beats arbitrary SSL: JEPA/VICReg/masked reconstruction used physically meaningless targets. Reprojection is geometrically grounded: "if your latent estimate is correct, you should predict what every electrode observes." Self-supervised but geometrically constrained.

7.3 Where the analogy breaks

The mapping is structural, not exact. Three important disanalogies:

1. **Each patient is a different "scene," not a different camera on the same scene.** In multi-view reconstruction, all cameras observe the SAME 3D scene simultaneously. Our patients are observed independently, at different times, producing different spatiotemporal patterns due to $r^{(i)}$. The shared structure f exists but is weaker than multi-view photometric consistency. The reprojection loss is therefore noisier—it must reconstruct through patient-specific residual, not just geometric projection.

2. **Reprojection is electrode-sparse, not pixel-dense.** A 256×256 image provides 65K reprojection constraints per view. Our 63–201 electrodes provide 100–1000× fewer constraints per frame. The reprojection signal per frame is thin; it aggregates value primarily across the temporal dimension ($T \approx 200$ frames/trial × 1315 trials).

3. **The "scene" changes every frame.** In static reconstruction, the scene is fixed across views. Here, $\mathbf{z}(t)$ changes continuously, so each frame is a different scene for reprojection. This is closer to **dynamic NeRF**—but the temporal structure is phoneme-specific and short-lived (∼1 s trials), not a persistent evolving scene.

These disanalogies mean: the reprojection loss will be noisier and less constraining than its vision analog; geometric inductive biases (distance bias, bilinear factorization) must compensate; and the reprojection loss is a useful regularizer but **should not be expected to carry the architecture** the way photometric loss carries NeRF. The supervised classification loss remains primary.

8 Validated Patterns from 3D Vision (2024–2026)

8.1 The field's trajectory: per-scene → feed-forward → foundation model

Our architecture sits at step 2. Key transferable patterns:

VGGT (CVPR 2025 Best Paper): joint 3D attribute prediction from variable-length views. Validates Perceiver jointly estimating latent state + alignment.

LVSM (ICLR 2025 Oral): encoder-decoder with fixed latent tokens vs. decoder-only. Decoder-only wins with enough data; explicit structure wins at small scale. *Validates our structured approach for* ∼1300 *trials.*

MMGN (ICLR 2024): continuous field reconstruction from sparse, irregularly sampled sensors using factorized spatiotemporal basis. **Most directly relevant.** Our bilinear decoder is structurally identical.

SPARF (CVPR 2023): NeRF from sparse and noisy poses. Directly analogous: our MNI coordinates are noisy poses; our electrodes are sparse views.

Senseiver (*Nature Machine Intelligence* 2023): Perceiver IO for field reconstruction from sparse sensors—structurally identical to our spatial encoder.

8.2 Architectural pattern validation

Our component	**Same pattern in vision**
Perceiver cross-attention (virtual → actual)	Canonical-to-observation tokens (LRM, LVSM)
Virtual electrodes at learnable positions	Triplane NeRF / fixed latent tokens
MNI distance-biased attention	Geometry-aware attention (GraphSplat, MVSplat)
Bilinear decoder $\hat{d} = \phi(\mathbf{p})^\top \psi(s)$	Spatiotemporal factorization (MMGN)
Reprojection loss	Photometric loss (all NeRF/3DGS)
Learnable per-patient Δ	Learnable camera extrinsics (SPARF, AnySplat)
Patient embedding $e^{(i)}$	Per-scene latent code (CodeNeRF)
Electrode masking	View dropout / masked pretraining (MAE)
Structured inductive bias for small data	Explicit geometry at small scale (LVSM)

8.3 What we are correctly NOT doing

Diffusion priors (need millions of scenes), 4D Gaussian primitives (trials are independent events), foundation model scale (1300 trials cannot support >200K params), decoder-only architecture (needs massive data), contrastive SSL (failed in our experiments).

8.4 The data scale gap

	3D Vision	**Our Problem**
Training scenes/patients	50K–1M+	∼20 patients
Observations per scene	10–100+ images (M pixels)	63–201 electrodes
Total data	TB scale	∼240 MB
Parameter budget	50M–500M+	∼200K

This gap means: we **must** use strong geometric inductive biases; the reprojection loss is valuable but electrode-sparse (Section 7.3); and our ∼200K architecture is correctly small. The field's finding that explicit geometric structure helps at small scale (LVSM, SPARF) directly validates our design.

9 Uncertainty Estimation

9.1 From binary to continuous confidence

The current pipeline has a binary mechanism: significant channel selection (keep/drop). Proper uncertainty replaces this with continuous per-electrode confidence.

9.2 Two types of uncertainty

Aleatoric (irreducible): "This electrode is inherently noisy." Per-electrode, does not decrease with more data.

Epistemic (reducible): "We lack data in this cortical region." Per-position, decreases with more patients.

9.3 Recommended uncertainty stack

9.3.1 Tier 1: Minimal implementation, high value

Beta-NLL loss (Seitzer et al., ICLR 2022). Replace MSE reprojection with learned per-electrode variance:

$$\mathcal{L}_{\text{reproj}}^{\text{UQ}} = \frac{1}{2}\Big(\log \hat{\sigma}_j^2 + \frac{(d_j - \hat{d}_j)^2}{\hat{\sigma}_j^2}\Big) \cdot \big(\hat{\sigma}_j^2\big)_{\text{sg}}^{0.5} \tag{7}$$

Scope: Beta-NLL affects the **training loss only**—it downweights reprojection gradients from noisy electrodes, preventing them from distorting the encoder. It does **not** directly modulate attention weights at inference time.

Learned confidence gate (for inference-time uncertainty weighting). To propagate aleatoric uncertainty into the Perceiver attention, we add a lightweight per-electrode confidence scalar:

$$c_j = \sigma\big(\text{Linear}(\text{token}_j)\big) \in (0, 1) \tag{8}$$

The confidence scales the attention **values** (not logits): $V_j \leftarrow c_j \cdot V_j$ before aggregation. High-impedance or low-SNR electrodes learn $c_j \approx 0$. This is architecturally separate from Beta-NLL and adds $d{+}1 = 65$ parameters.

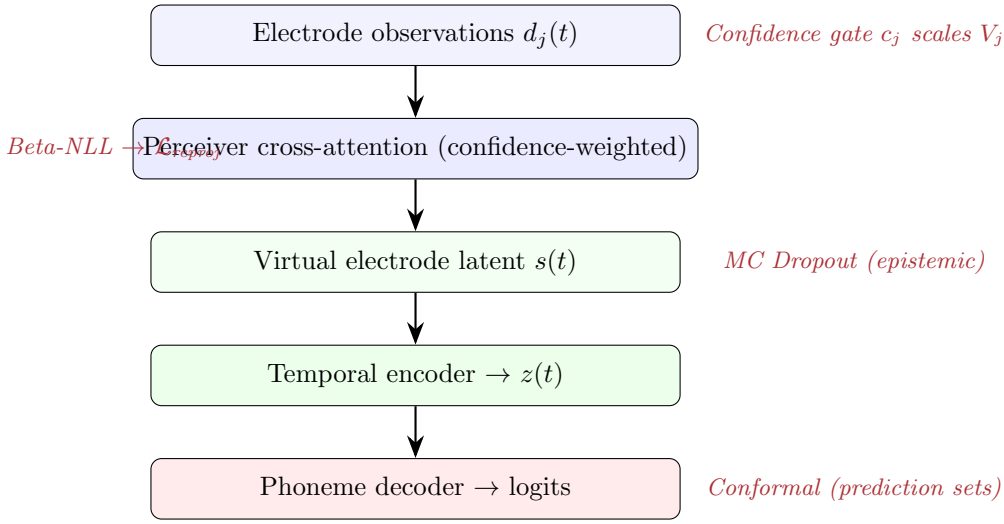
MC Dropout at inference (Gal & Ghahramani 2016). Keep dropout active, 10–20 passes, variance = epistemic uncertainty. Remove dropout from output layer.

Conformal prediction (Angelopoulos & Bates 2021). Post-hoc prediction **sets** with coverage guarantees. Use CV fold as calibration. Clinical value: "model thinks /b/ or /p/ (both labial stops)."

9.3.2 Tier 2: Moderate implementation

Evidential regression (NIG prior): single-pass decomposed aleatoric + epistemic per electrode. **Probabilistic virtual electrodes** (pixelSplat-style): predict latent distributions, uncertainty propagates through the pipeline.

9.4 How uncertainty actually connects to the architecture



Beta-NLL affects training gradients (left). Confidence gate c_j affects inference-time attention (right). MC Dropout captures model uncertainty. Conformal prediction gives calibrated prediction sets. Each mechanism has a distinct scope—they are complementary, not redundant.

Part III

Architecture Design

10 Design Principles

Every component maps to the observation function, with an explicit distinction between two latent representations:

Symbol	Meaning	Architectural component
$d_j^{(i)}(t)$	Electrode observation	Input token
$\mathbf{p}_j^{(i)}$	MNI position (approximate)	MNI positional encoding
$\Delta^{(i)}, \omega^{(i)}$	Registration correction (translation + rotation)	Learnable 6D offset
$s(t)$	**Per-frame spatial summary**—instantaneous field estimate	Spatial encoder output
$\mathbf{z}(t)$	**Temporally-refined state**—with sequence context	Temporal encoder output
$e^{(i)}$	Patient residual	Learnable embedding
$g(\cdot)$	Shared somatotopic map	Observation decoder

The $s(t)$ vs. $\mathbf{z}(t)$ distinction. The theory posits a single latent state $\mathbf{z}(t)$. The architecture produces two representations at different processing stages:

- $s(t) \in \mathbb{R}^{L \times d}$: the spatial encoder output. Per-frame, no temporal context. Captures the instantaneous spatial pattern of cortical activity.
- $\mathbf{z}(t) \in \mathbb{R}^{d}$: the temporal encoder output. Incorporates multi-scale temporal context from neighboring frames.

The observation decoder uses $s(t)$ for reprojection because it reconstructs a per-frame spatial pattern. The phoneme decoder uses $\mathbf{z}(t)$ because phoneme classification requires temporal dynamics. These serve complementary training objectives: $\mathcal{L}_{\text{reproj}}$ trains the spatial encoder to faithfully capture the cortical field per frame, while $\mathcal{L}_{\text{class}}$ trains the temporal encoder to extract phoneme-discriminative trajectories. $s(t)$ is **not** the articulatory state $\mathbf{z}$ from Eq. (4)—it is the architecture's per-frame estimate of the spatial field, from which $\mathbf{z}(t)$ is later refined.

Objective tension. $\mathcal{L}_{\text{reproj}}$ wants $s(t)$ to preserve all variability (for faithful reconstruction); $\mathcal{L}_{\text{class}}$ wants $s(t)$ to be discriminative (discarding irrelevant variance). These objectives share the spatial encoder and can compete. The temporal encoder provides a buffer, and λ annealing ($0.5 \to 0.1$) gradually prioritizes classification. If the tension is severe, λ=0 (pure supervised) is the fallback (tested in ablation A5).

11 Spatial Encoder (∼100K params)

11.1 Electrode tokens

For electrode j on patient i at time t:

$$\text{token}_j = \underbrace{\text{Linear}(d_j \to d)}_{\text{activity}} + \underbrace{\text{Linear}(\tilde{\mathbf{p}}_j \to d)}_{\text{MNI PE}} + \underbrace{\text{Linear}(\text{row}_j, \text{col}_j \to d)}_{\text{Grid PE}} + \underbrace{e^{(i)}}_{\text{patient}} \tag{9}$$

where $\tilde{\mathbf{p}}_j = R(\omega^{(i)}) \cdot \mathbf{p}_j + \Delta^{(i)}$ is the corrected MNI position.

Per-patient geometric parameters ($6 + d$ per patient):

- $\Delta^{(i)} \in \mathbb{R}^3$: learnable translation, initialized to $\mathbf{0}$.
- $\omega^{(i)} \in \mathbb{R}^3$: learnable rotation (axis-angle $\to$ rotation matrix via Rodrigues), initialized to $\mathbf{0}$. Corrects array rotation/orientation error.
- $e^{(i)} \in \mathbb{R}^d$: learnable patient embedding, initialized $\sim \mathcal{N}(0, 0.02)$.

Assumption: The patient embedding $e^{(i)}$ is added uniformly to all electrode tokens. This means the encoder cannot apply **position-dependent** patient-specific corrections—only the observation decoder can (via $\text{MLP}_{\text{spatial}}(\tilde{\mathbf{p}}, e)$). For tumor patients with spatially non-uniform geometric distortion $\delta_i(\mathbf{p})$, this is a known limitation. The rotation $\omega^{(i)}$ partially addresses this (correcting rigid misalignment beyond translation), but cannot capture nonlinear warping. Ablation A3 tests whether richer per-patient parameterization (e.g., position-dependent $e^{(i)}$) helps.

Dead electrodes are excluded (not masked). Variable N_i is handled natively by attention.

11.2 Perceiver cross-attention with distance bias

$L = 16$ **virtual electrodes** with learnable MNI positions, initialized to cover the convex hull of source electrode positions:

$$\text{query}_i = \text{embed}_i + \text{MNI_PE}(\text{v_pos}_i) \tag{10}$$

Cross-attention with spatial locality prior:

$$\boxed{A_{ij} = \frac{Q_i \cdot K_j}{\sqrt{d}} \; - \; \alpha \cdot \|\text{v_pos}_i - \tilde{\mathbf{p}}_j\|^2} \tag{11}$$

where $\alpha > 0$ is learnable (initialized 0.1), controlling spatial blur.

Attention values are scaled by the learned confidence gate (Section 9):

$$\text{out}_i = \sum_j \text{softmax}(A)_{ij} \cdot \big(c_j \cdot V_j\big) \tag{12}$$

where $c_j = \sigma(\text{Linear}(\text{token}_j)) \in (0, 1)$. High-noise electrodes contribute less.

4-head attention, optional 1-layer self-attention (captures inter-zone co-articulation).

11.3 Spatial readout: attention pooling (not MaxPool)

The $L = 16$ virtual electrode outputs form a **spatially-structured representation** (L, d) where each virtual electrode captures activity from a different cortical zone. Naïve MaxPool would discard which zone contributed what—destroying the spatial decomposition the virtual electrodes were designed to provide.

Instead, we use **attention-based readout**:

$$s_{\text{pool}}(t) = \sum_{i=1}^{L} w_i \cdot \text{out}_i(t), \qquad w_i = \text{softmax}\big(\mathbf{q}_{\text{read}}^\top \text{out}_i(t)/\sqrt{d}\big) \tag{13}$$

where $\mathbf{q}_{\text{read}} \in \mathbb{R}^d$ is a single learnable readout query. This is a soft selection over zones—preserving the somatotopic decomposition through learned weighting. Unlike MaxPool, the model can learn to weight tongue-zone and lip-zone outputs differently depending on the current activity pattern.

Full spatial output: $s(t) \in \mathbb{R}^{L \times d}$ (all virtual electrode outputs) is preserved for the observation decoder and optionally for the temporal encoder (ablation A8). $s_{\text{pool}}(t) \in \mathbb{R}^d$ is the default input to the temporal encoder.

11.4 Parameter count

Component	**Params**
Activity embedding, MNI PE, Grid PE	576
Virtual embeddings + positions	1,072
Cross-attention + self-attention + FFN	∼98K
Confidence gate	65
Readout query	64
Distance scale α	1
Total spatial encoder	**∼100K**
Per-patient $(\Delta + \omega + e)$	$6 + d$/patient

12 Observation Decoder (∼25K params)

The forward observation model—the "renderer" in the multi-view analogy. Uses the full spatial output $s(t) \in \mathbb{R}^{L \times d}$:

$$\boxed{\hat{d}_j(t) = \phi(\tilde{\mathbf{p}}_j,\ e^{(i)})^\top\ \psi\big(\bar{s}(t)\big)} \tag{14}$$

where $\bar{s}(t) = \text{mean}(s(t)) \in \mathbb{R}^d$ is the mean-pooled spatial summary, and:

- $\phi = \text{MLP}_{\text{spatial}}(\tilde{\mathbf{p}}, e) \in \mathbb{R}^k$: spatial tuning vector at corrected position.
- $\psi = \text{MLP}_{\text{latent}}(\bar{s}) \in \mathbb{R}^k$: articulatory activation for current state estimate.
- $k = 32$: bilinear interaction dimensionality.

Note: The decoder takes $\bar{s}(t)$, not $\mathbf{z}(t)$. Since reprojection is a per-frame spatial task (reconstructing electrode observations from the instantaneous field estimate), temporal context is unnecessary. The temporal encoder is trained only through $\mathcal{L}_{\text{class}}$.

Assumption: The bilinear form $\hat{d} = \phi(\mathbf{p})^\top \psi(s)$ assumes the observation function **separates** into spatial tuning and articulatory activation interacting through an inner product. This means: the response of an electrode at position $\mathbf{p}$ to state s is a linear combination of k independent modes. Higher-order interactions (e.g., coarticulation effects where tongue position modulates lip-area neural activity) cannot be captured. This is a strong structural assumption, validated by MMGN for physical fields but unverified for coarticulatory neural dynamics. Ablation A9 tests a full MLP fallback $\hat{d} = \text{MLP}([\tilde{\mathbf{p}}; e; \bar{s}])$ at the cost of losing the factorization's interpretability and inductive bias.

Position-dependent patient correction. Unlike the encoder (where $e^{(i)}$ is a global additive bias), the decoder's $\text{MLP}_{\text{spatial}}(\tilde{\mathbf{p}}, e)$ can learn position-dependent patient effects: "for this patient, electrodes near position $\mathbf{p}$ respond differently." This provides the capacity to partially model the nonlinear residual $\delta_i(\mathbf{p})$ that the global $\Delta^{(i)}, \omega^{(i)}$ cannot.

Reprojection loss (self-supervised):

$$\mathcal{L}_{\text{reproj}} = \frac{1}{N_i} \sum_{j=1}^{N_i} \left\| d_j(t) - \hat{d}_j(t) \right\|^2 \tag{15}$$

Electrode masking: Drop 20–40% of electrodes from encoder input; keep them in reprojection loss. The model must spatially interpolate—geometric masked reconstruction.

13 Temporal Encoder (∼70K params)

13.1 Multi-scale tokenization

$$\begin{aligned} s_{\text{pool}}(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=3,s=3)} T_3 \approx 67 \text{ tokens at } \sim 67\,\text{Hz} \\ s_{\text{pool}}(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=5,s=5)} T_5 \approx 40 \text{ tokens at } \sim 40\,\text{Hz} \\ s_{\text{pool}}(1{:}T) &\xrightarrow{\text{Conv1d}(d,d,k=10,s=10)} T_{10} \approx 20 \text{ tokens at } \sim 20\,\text{Hz} \end{aligned} \tag{16}$$

All ≈127 tokens concatenated with learnable per-scale embeddings + sinusoidal temporal PE.

Caveat: Multi-scale stride 3+5+10 was validated on BiGRU. The temporal transformer processes sequences differently (parallel attention vs. recurrent). The articulatory timescales are physics-based (not architecture-specific), so the stride values should be approximately correct, but the optimal configuration should be re-validated.

13.2 Temporal transformer

2 layers, 4 heads, d=64, FFN dim 256. Pre-norm transformer. Each token attends across all scales and times.

13.3 Readout

$z_{\text{global}} = \text{mean_pool}(\text{all tokens}) \in \mathbb{R}^d$ for k-NN. $z_{\text{pos}} = \text{mean_pool}(\text{tokens in phoneme window}) \in \mathbb{R}^d$ per position for classification.

14 Phoneme Decoder (∼5K params)

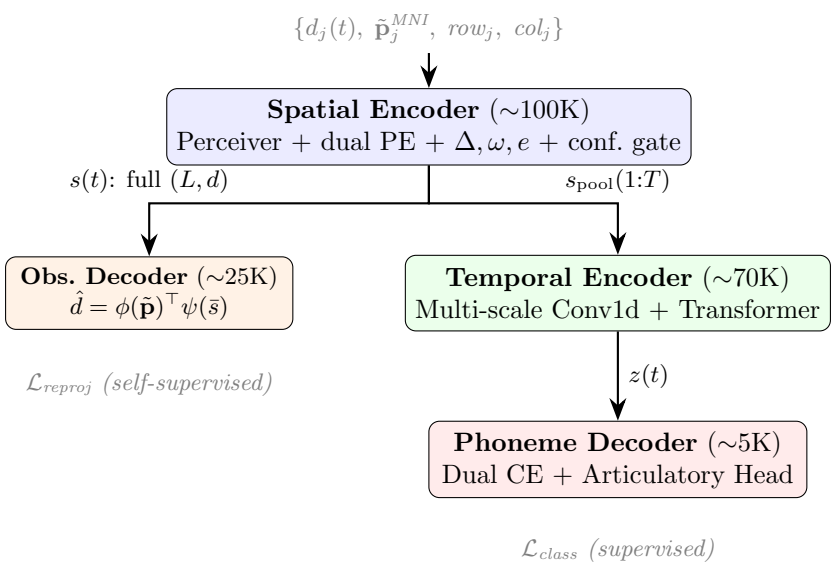
Dual head:

$$\text{logits}_p = \frac{1}{2}\Big(\underbrace{\text{Linear}(z \to 9)}_{\text{CE head}} + \underbrace{\tau \cdot \text{normalize}(\text{Linear}(z \to 15)) \cdot A^\top}_{\text{articulatory head}}\Big) \tag{17}$$

where $A \in \mathbb{R}^{9\times 15}$ is the fixed articulatory feature matrix and τ is learnable temperature.

$\mathcal{L}_{\text{class}} = \frac{1}{3}\sum_{p=1}^{3} \text{FocalCE}(\text{logits}_p, y_p;\ \gamma{=}2, \text{LS}{=}0.1)$ with mixup (α=0.2).

15 Complete Architecture



$$\boxed{\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \cdot \mathcal{L}_{\text{reproj}}} \tag{18}$$

Module	Params	Role
Spatial encoder	∼100K	Invert observation function
Observation decoder	∼25K	Forward model (reprojection)
Temporal encoder	∼70K	Dynamics modeling
Phoneme decoder	∼5K	Classification
Total shared	**∼200K**	
Per-patient (×10)	700	Registration + identity
Grand total	**∼201K**	(cf. current pipeline: 122K)

Part IV

Training, Evaluation, and Experimental Design

16 Training Pipeline

16.1 Phase 1: Source patient training (supervised + reprojection)

9 source patients, ∼1315 labeled trials. Each batch:

1. **Spatial encoding**: construct tokens, mask 20–40% electrodes, Perceiver → $s(t)$, attention pooling → $s_{\text{pool}}(t)$.
2. **Reprojection**: decoder predicts all electrodes (including masked) from $\bar{s}(t) = \text{mean}(s(t))$; compute $\mathcal{L}_{\text{reproj}}$ (or $\mathcal{L}_{\text{reproj}}^{\text{UQ}}$).
3. **Temporal encoding**: $s_{\text{pool}}(1{:}T) \to$ multi-scale → transformer → $z(t)$.
4. **Classification**: dual head → $\mathcal{L}_{\text{class}}$.
5. **Total**: $\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \mathcal{L}_{\text{reproj}}$ (λ: 0.5 → 0.1 annealed).

Optimizer: AdamW with differential LR:

- Shared parameters: 10^{-3}
- Virtual positions, Δ, ω: 3×10^{-4} (slow—geometric stability)
- Patient e: 10^{-3}

Linear warmup (20 epochs) → cosine decay. Weight decay 10^{-4}. Gradient clipping 5.0. Batch size 16. Early stopping on held-out 20% source validation, patience 7.

16.2 Phase 2: Target adaptation (LP-FT)

Per CV fold on target patient (5-fold grouped-by-token):

1. **Linear probe**: freeze encoder, train decoder + $\Delta^{(\text{target})} + \omega^{(\text{target})} + e^{(\text{target})}$. Initialize $e = \text{mean}(e^{(\text{source})})$. 150 epochs. Loss: $\mathcal{L}_{\text{class}} + 0.1 \cdot \mathcal{L}_{\text{reproj}}$.
2. **Fine-tune**: unfreeze encoder at 0.1× LR. 100 epochs, patience 7.

16.3 Phase 3 (optional): Reprojection pre-training

Pre-train spatial encoder + observation decoder on all available data using $\mathcal{L}_{\text{reproj}}$ only. No labels needed. Then initialize Phase 1 from pre-trained weights.

17 Evaluation Protocol

PER via edit distance, grouped-by-token 5-fold CV. Full recipe: weighted k-NN (k=10, cosine), TTA 16, dual head, best-of per fold, source k-NN at 0.5× weight. Conformal prediction sets as interpretability output.

18 Ablation Experiments

ID	Ablation	Question
A1	MNI PE present/absent	**Central**: do MNI coordinates help?
	a. Current Conv2d + BiGRU baseline	→ 0.762
	b. Perceiver + Grid PE only	→ architecture benefit
	c. Perceiver + MNI + Grid PE	→ MNI benefit
	d. (c) + reprojection loss	→ reprojection benefit
A2	**Population-level (all patients)**	**Is the 0.76 wall S14-specific?**
A3	Per-patient params	Δ only / $+\omega$ / $+e$ / position-dep. e
A4	Virtual electrode count	$L \in \{4, 8, 16, 32\}$
A5	Reprojection weight	$\lambda \in \{0, 0.01, 0.1, 0.5\}$
A6	Cross-task pooling	+ Duraivel data (architecture-independent)
A7	Uncertainty stack	Beta-NLL + conf. gate vs. MSE-only
A8	Spatial readout	MaxPool / attention pooling / full (L, d) to temporal
A9	Bilinear vs. MLP decoder	$\phi^\top \psi$ vs. MLP($[\tilde{\mathbf{p}}; e; \bar{s}]$)
A10	Temporal backbone	Transformer vs. BiGRU (+ stride re-validation)

19 Experimental Priorities

The document's own evidence suggests a clear priority ordering that should be stated transparently:

1. **Cross-task data pooling** (A6). Lowest risk, highest confidence. Same phonemes, overlapping patients, validated by Duraivel 2025. Directly addresses target-data scarcity. Can be tested with the current pipeline **and** the new architecture. Should be pursued first, in parallel with coordinate sourcing.

2. **Population-level evaluation** (A2). Pure methodology fix. Required to determine whether the 0.76 wall is S14-specific (55-experiment overfitting) or universal. Must be the **primary success criterion**, not just an ablation. Zero architectural risk.

3. **MNI coordinate-based architecture** (A1, this document). A working hypothesis. The evidence that spatial alignment is THE binding constraint is suggestive but unvalidated: it has not been isolated from data scarcity, measurement noise, or S14-specific effects (Section 5.4). The architectural investment is justified by: (a) MNI coordinates are currently unused information; (b) the observation function formulation provides a principled framework; (c) every component

has a validated analog in 3D vision. But the hypothesis could be wrong—coordinates may not help if the wall is data-limited rather than alignment-limited.

The architecture is the most speculative of the three paths. This is acceptable—it is the path with the most intellectual contribution if it works—but the document should not obscure the priority ordering.

20 Success Criteria

Primary criterion (population-level):

- Mean PER across all patients improves over current LOPO baseline with best architecture (A1c/d vs. A1a), evaluated across $\geq$3 seeds.
- This is the only criterion that provides scientific evidence for the MNI coordinate hypothesis.

Secondary criteria (S14 pilot—informative but not definitive):

- A1c outperforms A1b by $\geq$2 pp on S14 (MNI adds value beyond architecture change).
- PER < 0.740 on S14 (LOPO exceeds per-patient).

Caveat: S14 criteria are contaminated by 55+ prior experiments. Population-level results should take precedence if they disagree.

Interpretability (qualitative validation):

- Virtual electrode positions cluster over known speech motor regions.
- Learned $\Delta^{(i)}, \omega^{(i)}$ correlate with known registration quality (larger for tumor patients).
- Spatial tuning vectors $\phi(\mathbf{p})$ recover articulatory somatotopy when visualized.
- Confidence gate c_j correlates with electrode impedance or SNR.

21 Available Data

Data source	Volume	Usage	Risk
PS task HGA	9 pts, $\sim$1315 trials	Source training	–
S14 HGA	153 trials	Target evaluation	–
Cross-task (Duraivel)	3 pts, 156–208 trials	**Priority 1: more data**	Low
MNI coordinates	All patients	Perceiver architecture	Medium
Continuous recordings	$\sim$168 min	Phase 3 reprojection SSL	Medium
Lexical task patients	13 patients	Additional sources	Medium

22 MNI Coordinate Sourcing

Hard dependency. Likely locations: BioImage Suite outputs (DBS), Brainlab (tumor), ECoG_Recon on Box. Fallback: scaled ACPC grid positions (retains within-patient structure, loses cross-patient alignment).

Risk	Sev.	Mitigation
MNI coordinates unavailable	High	Fallback to Grid PE only
MNI alignment too imprecise	High	Ablation A1; learnable α adapts blur
Reprojection competes with classification	Medium	λ ablation A5; λ=0 fallback
Perceiver overfits (200K, 1300 trials)	Medium	Masking; early stopping; reprojection reg.
Bilinear separability too restrictive	Medium	Ablation A9: MLP fallback
Per-patient params insufficient for tumor distortion	Medium	Ablation A3; position-dep. e option
Virtual electrodes collapse	Low	Initialize across convex hull; monitor

23 Risk Assessment

24 Summary

This document formalizes cross-patient speech decoding as **multi-view neural field reconstruction** with explicit acknowledgment of where the analogy holds and where it breaks (Section 7.3).

The architecture produces two distinct representations: $s(t)$ (per-frame spatial summary, trained via reprojection) and $\mathbf{z}(t)$ (temporally-refined state, trained via classification). The observation decoder uses a bilinear factorization that assumes separability between spatial tuning and articulatory activation—a structural hypothesis validated in physical field reconstruction but unverified for coarticulatory neural dynamics (ablation A9).

Key improvements over v2: (1) explicit $s(t)$ vs. $\mathbf{z}(t)$ distinction resolving the theory-architecture mismatch; (2) attention-based readout replacing MaxPool to preserve somatotopic decomposition; (3) per-patient rotation $\omega^{(i)}$ alongside translation $\Delta^{(i)}$; (4) mechanically specified uncertainty connection (confidence gate c_j on attention values, Beta-NLL on reprojection loss); (5) honest acknowledgment of the bilinear separability assumption; (6) "where the analogy breaks" section; (7) population-level as primary success criterion; (8) transparent experimental priority ordering (data > population eval > architecture); (9) expanded ablation plan (A1–A10); (10) flagged assumptions in red throughout.

Total budget: ∼200K shared + 70 per patient. Every component maps to the observation equation, with structural assumptions explicitly marked for empirical validation.